

Scalability & Future Plan

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Team ID: XXXXXX

Project Name: Human Development Index (HDI) Prediction using Machine Learning

Scalability & Future Plan

This document describes how the **Human Development Index (HDI) Prediction System** can be expanded to support more users, larger datasets, and advanced features in the future. It also outlines the team's roadmap for continuously improving the system to meet future requirements and enhance decision-making capabilities.

Current System Limitations

S.No	Limitation	Impact	Priority to Address
1	The system uses a static HDI dataset and does not update automatically.	Prediction accuracy depends on manually updated data.	High
2	The application is designed for a limited number of concurrent users.	Performance may decrease with heavy traffic.	Medium
3	The system predicts only the HDI category and provides limited recommendations.	Users receive limited decision-support insights.	Medium

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports small to medium number of users.	Deploy on cloud platforms (AWS, Azure, Render) with auto-scaling support.
2	Data Storage	Stores data in a local MySQL database.	Use cloud databases such as MySQL Cloud or PostgreSQL for larger datasets.
3	Performance	Single-server Flask application.	Implement caching, load balancing, and optimized machine learning models for faster predictions.
4	Security	Basic user authentication.	Add multi-factor authentication, role-based access control, and encrypted data transmission using HTTPS.

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Deploy the application on a cloud platform.	3 Months	Improve accessibility and support more users.
Phase 2	Integrate real-time HDI datasets from trusted sources such as UNDP and the World Bank.	6 Months	Ensure up-to-date predictions using the latest development data.
Phase 3	Add AI-based recommendations for improving healthcare, education, and income indicators.	9 Months	Provide personalized policy suggestions for better decision-making.
Phase 4	Develop a mobile application and multilingual interface.	12 Months	Increase usability and reach a wider audience across different regions.

Future Enhancement Summary

The **Human Development Index (HDI) Prediction System** is designed to be scalable and adaptable to future needs. By adopting cloud deployment, real-time data integration, enhanced security, and AI-powered recommendations, the system can support a larger user base while providing more accurate and actionable insights. Future developments such as

mobile access and multilingual support will further improve accessibility, making the platform a valuable tool for governments, researchers, educational institutions, and international development organizations.